

Interpretable and generative models for mouse spaceflight

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Biological & Physical Sciences

National Aeronautics and
Space Administration



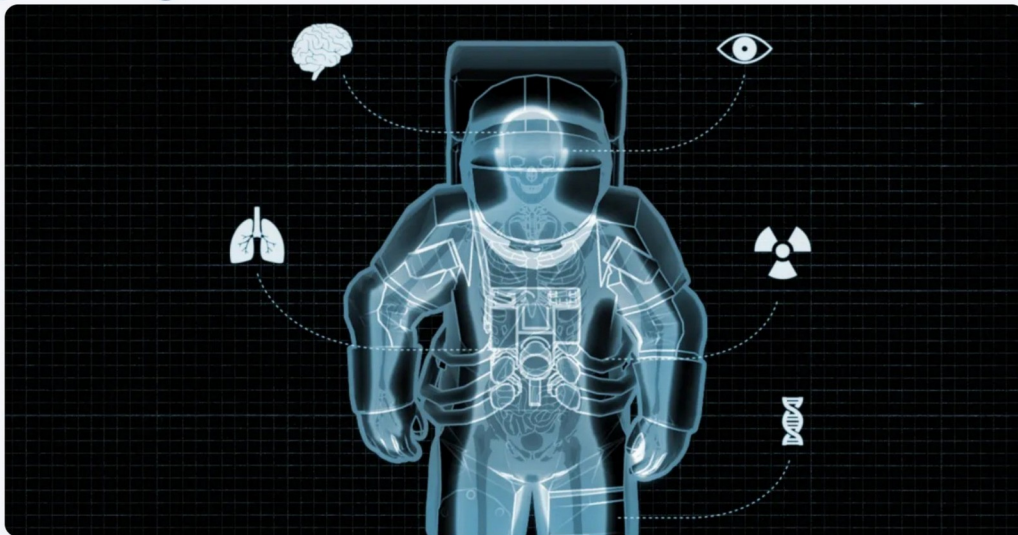
PROJECT OBJECTIVE

Learn how spaceflight changes living systems, then model the biology it produces

Using mouse bulk RNA-seq from NASA missions, we compare spaceflight (FLT) and ground-control (GC) mice across many tissues.

GOAL 1 · UNDERSTAND

Which genes, pathways, and systems does spaceflight alter?

**GOAL 2 · MODEL & GENERATE**

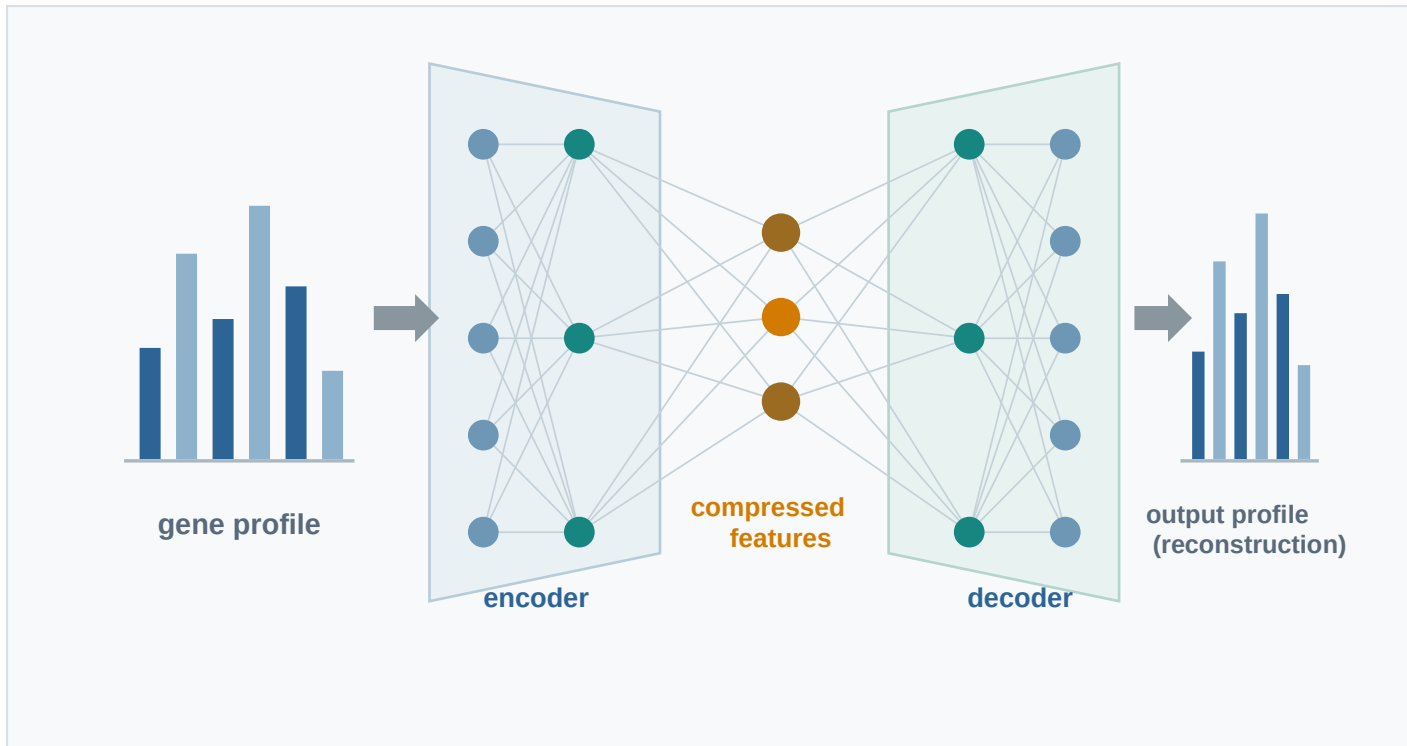
Learn meaningful representations and generate realistic profiles



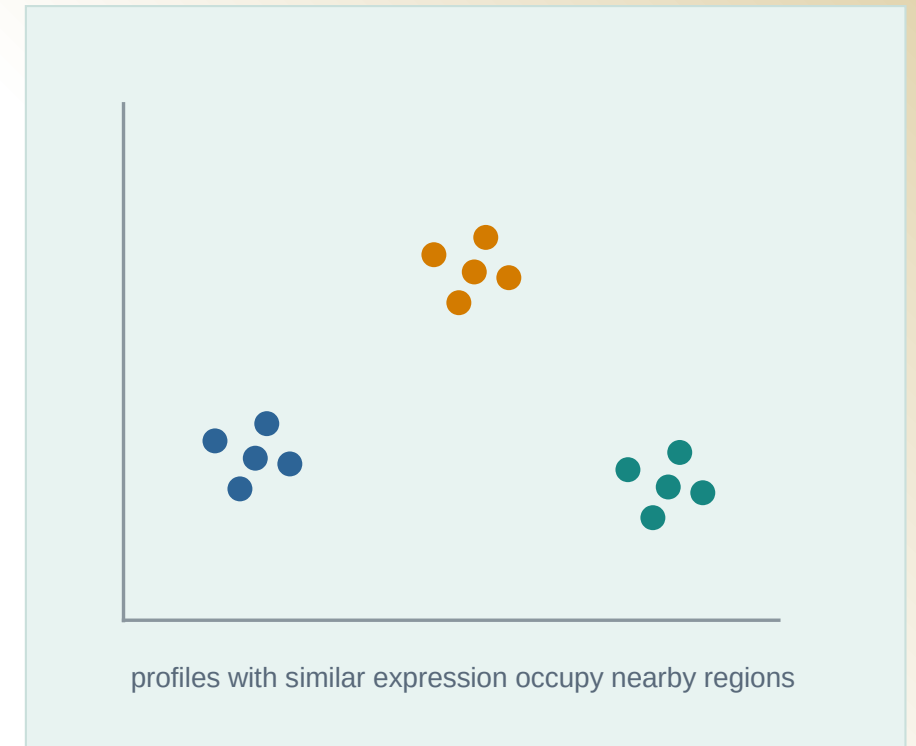
Autoencoders compress thousands of genes into a few features

The encoder summarizes a gene profile; the decoder learns enough structure to reconstruct it.

AUTOENCODER



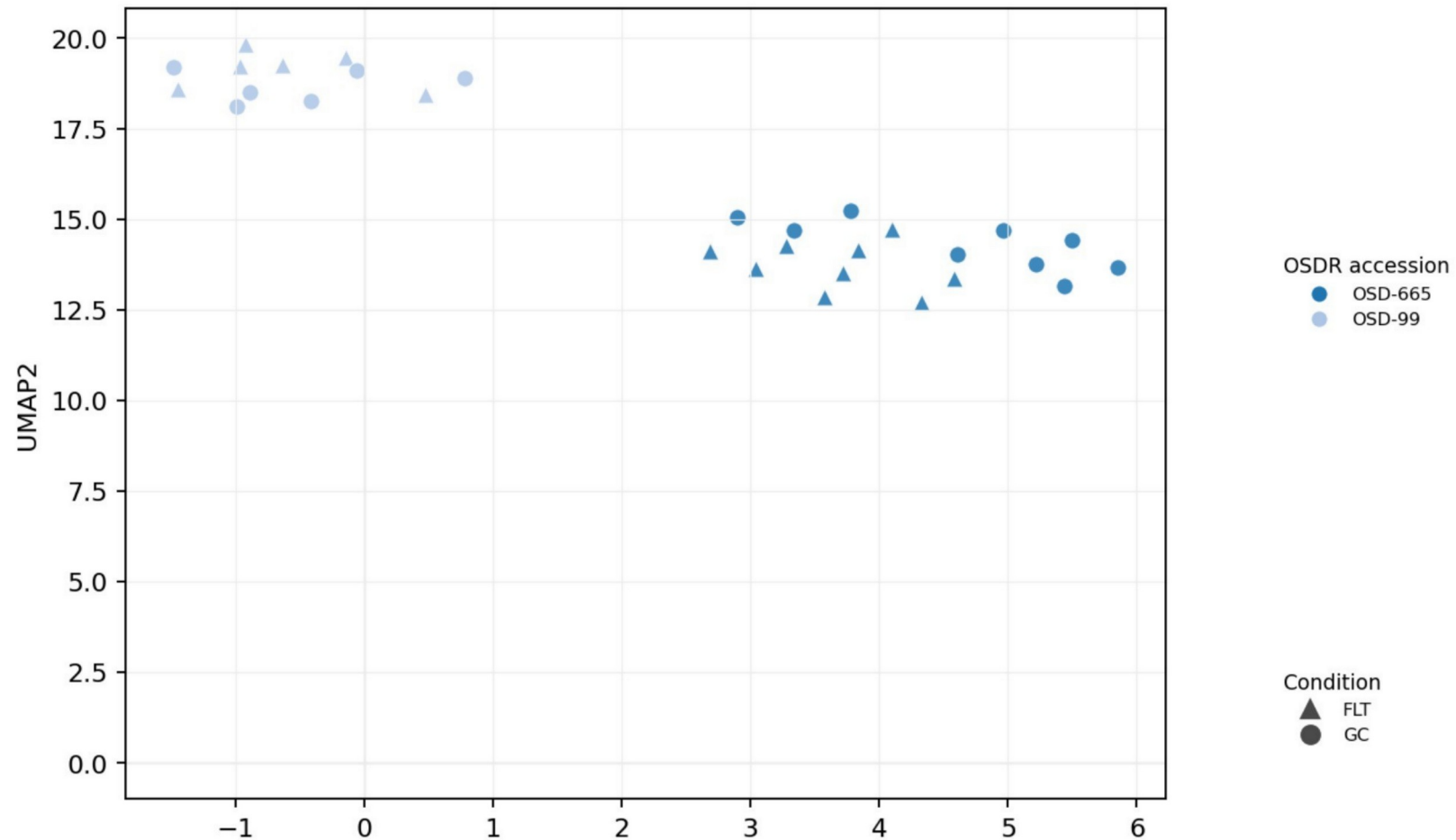
COMPRESSED SPACE



THE COMPLICATION

Study identity dominates the expression structure

In EDL muscle, the two studies separate while FLT and GC overlap within each study.



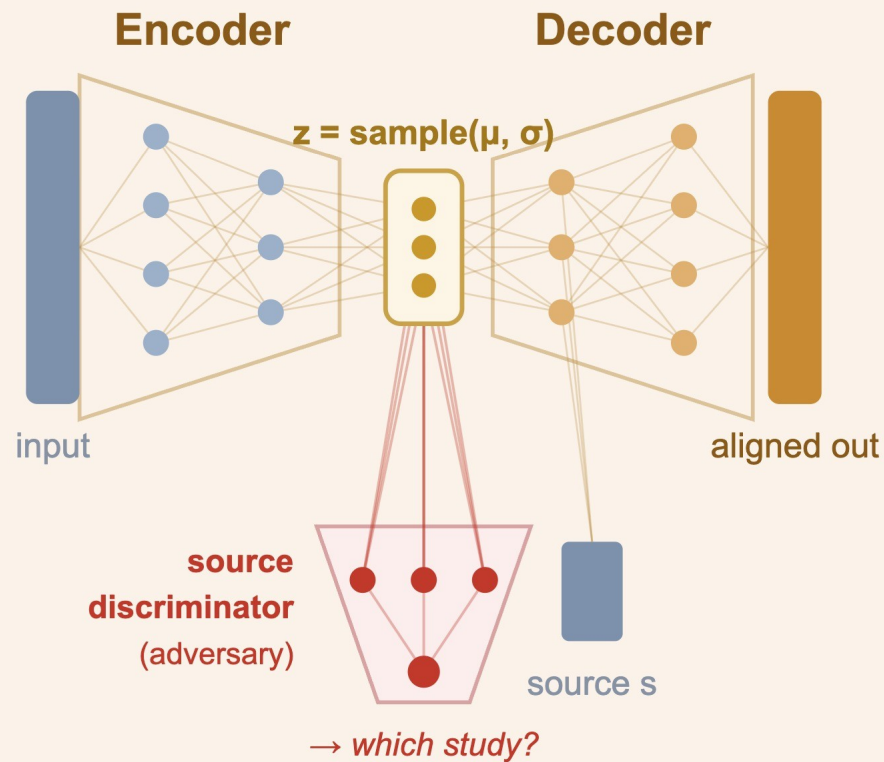
THE MAIN CHALLENGE

Can we just subtract the study effect?

MOBER (Multi-Origin Batch Effect Remover) is a deep domain-adaptation model that pulls studies into one distribution while trying to preserve biology.

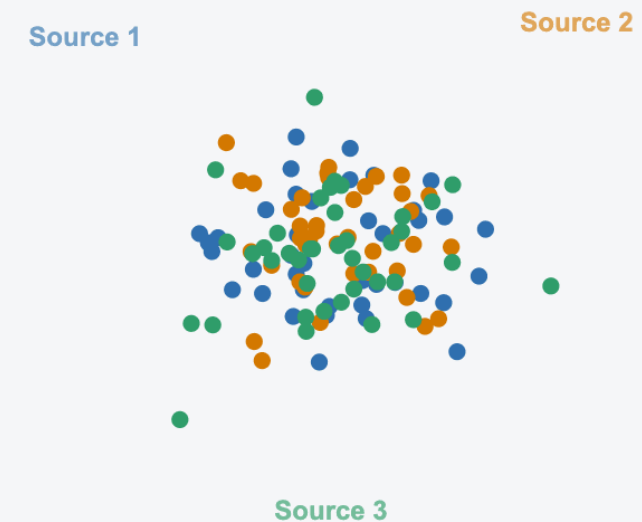
How MOBER works

VAE encoder-decoder + source-discriminator adversary



What it does · align studies

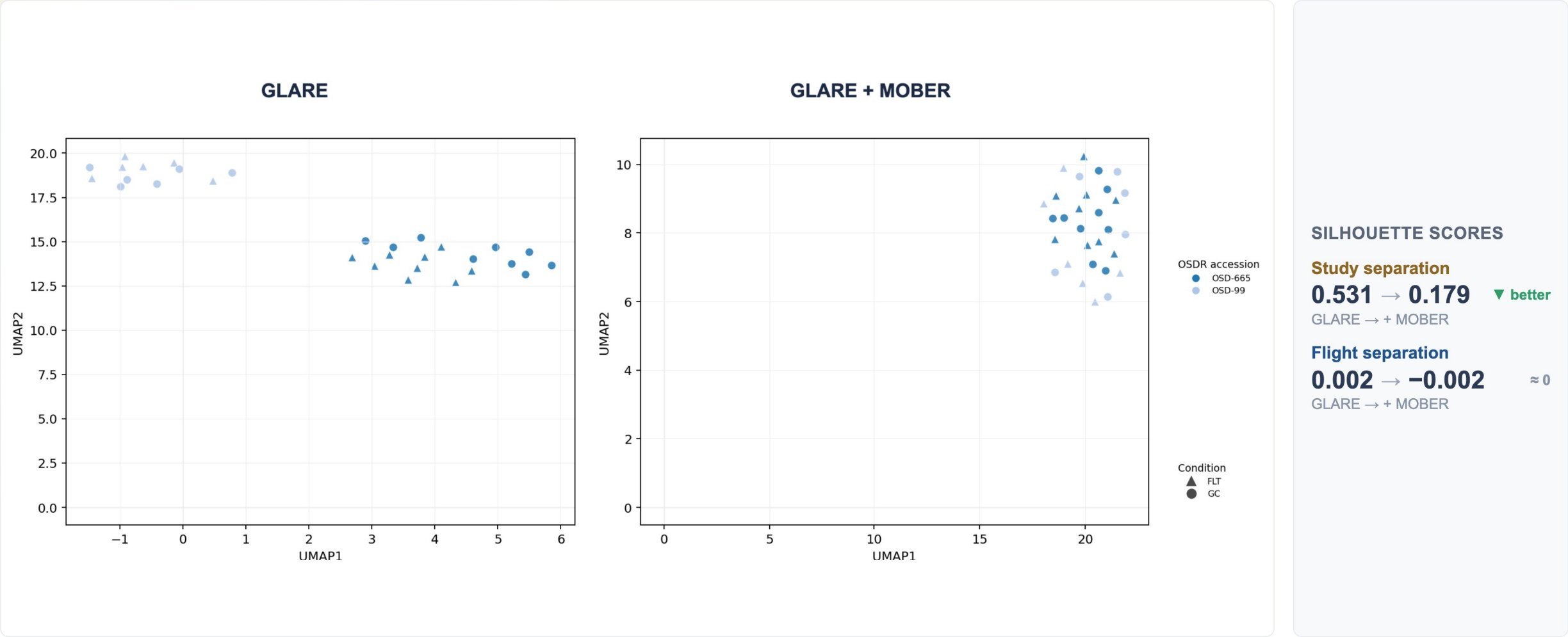
pull separate studies into one shared distribution



THE MAIN CHALLENGE · EVIDENCE

MOBER merges the two muscle studies, but flight and ground still overlap

Skeletal muscle (EDL): here MOBER **does** pull the two studies together, but FLT vs GC still doesn't separate, and the correction is only partial.



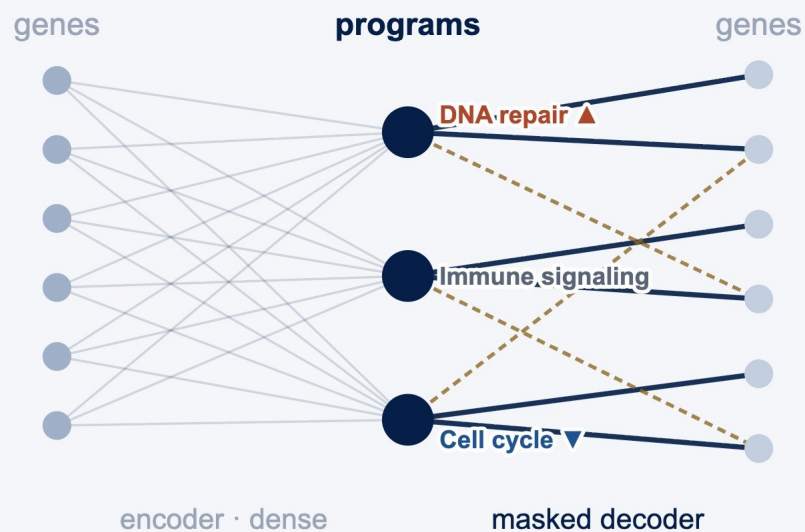
METHOD SPOTLIGHT

Don't learn the biology, wire it in

GLARE and MOBER tried to **learn** structure from the data. **expiMap** flips it: known Reactome pathways are **built into** the decoder, so each latent axis **is** biology by construction.

expiMap

STANDARD ENCODER · PATHWAY-MASKED DECODER

**WHY IT'S INTERPRETABLE**

- Each latent node is one Reactome program (~1,140).
- A soft mask lets a stray gene help only when the data support it.
- Compare FLT and GC one program at a time.

EXPIMAP SCORES

Program scores summarize pathway changes within one tissue

	FLT score	-	GC score	=	FLT - GC
Extracellular matrix	+0.47	-	+0.08	=	+0.39
Cell-cycle control	+0.42	-	+0.10	=	+0.32
Immune signaling	+0.15	-	+0.13	=	+0.02
Oxidative metabolism	-0.06	-	-0.03	=	-0.03
DNA repair	-0.10	-	+0.34	=	-0.44

■ lower ■ little change ■ higher

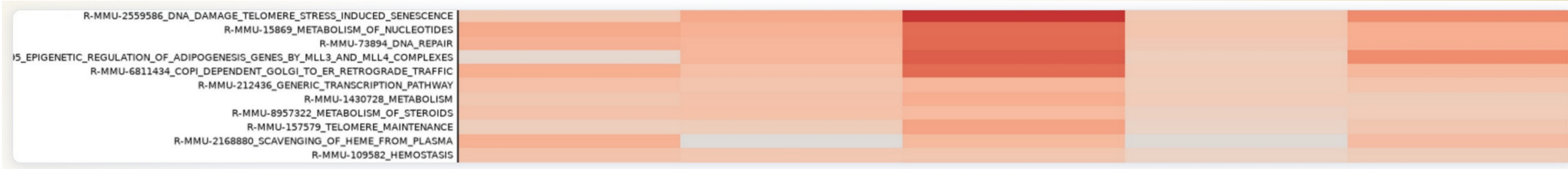
OUTPUT · PROGRAMS × SAMPLES



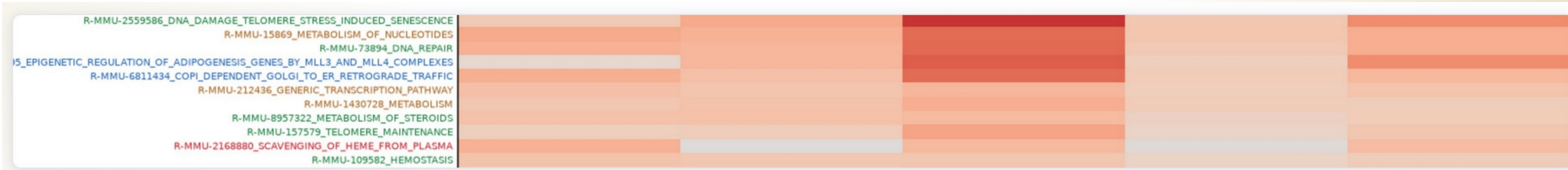
Each program, reviewed against prior literature

The pathway scores stay fixed; the pathway-name colors summarize the literature comparison.

1 PATHWAY SCORES



2 LITERATURE LABELS



Aligned Complementary Uncertain Conflicting Low effect

Name color records the literature assessment; heatmap color records the program score.

EXPIMAP RESULTS

Five tissues showed recurring pathway patterns

FLT and GC were compared within each study before the study-level changes were combined.

Thymus

117 samples | 5 projects

LOWER IN FLIGHT

DNA repair; RHOA cytoskeletal cycle; lymphoid-stromal interactions

Known involution may also involve lower repair, motility, and niche coordination.

Skin

151 samples | 4 projects

LOWER IN FLIGHT

Chromatin regulation; DNA repair; Hedgehog; sphingolipids; cell junctions

Barrier injury may include a broader loss of tissue maintenance and coordination.

Liver

197 samples | 9 projects

LOWER IN FLIGHT

MHC class II antigen presentation; T-cell receptor signaling

Metabolic heterogeneity coexisted with lower adaptive immune communication.

Spleen

100 samples | 5 projects

LOWER IN FLIGHT

T-cell receptor signaling; neutrophil degranulation; C-type lectin signaling

The most consistent multi-pathway result joined adaptive and innate immune changes.

Kidney

135 samples | 6 projects

HIGHER IN FLIGHT

ECM proteoglycans; WNT signaling; IGF transport and uptake

Matrix and growth-factor programs rose together, suggesting a renal remodeling response.

BACKGROUND

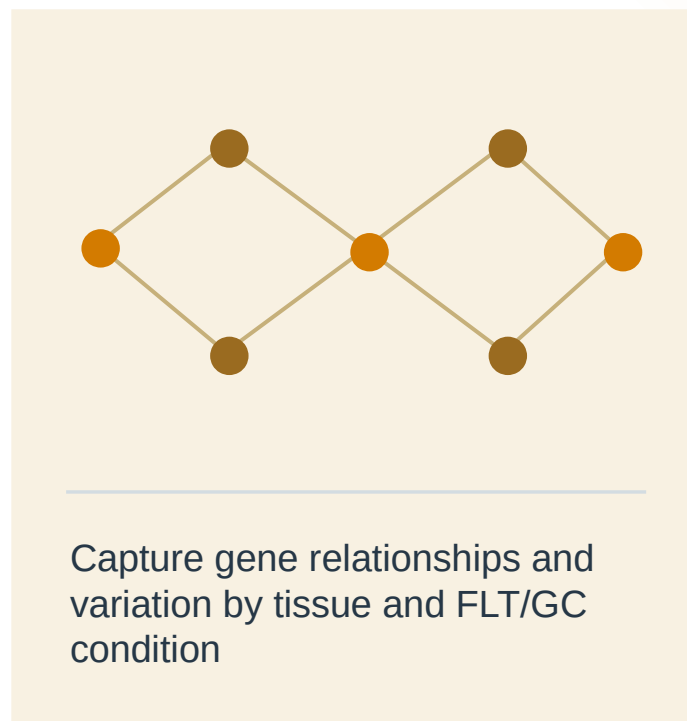
What is synthetic transcriptomics?

A generator learns patterns in measured RNA-seq data, then samples new expression vectors under chosen conditions.

01 Observed profiles



02 Learn expression patterns



03 Synthetic profiles



HOW THE GENERATORS WORK · 1 OF 2

Diffusion: denoise from noise

Start from pure noise and denoise step by step onto the data manifold, conditioned on FLT or GC.

FAMILIAR · IMAGE DIFFUSION



OURS · GENE-EXPRESSION DIFFUSION

reverse diffusion

noise

step $t = 1000$ 

● FLT ● GC

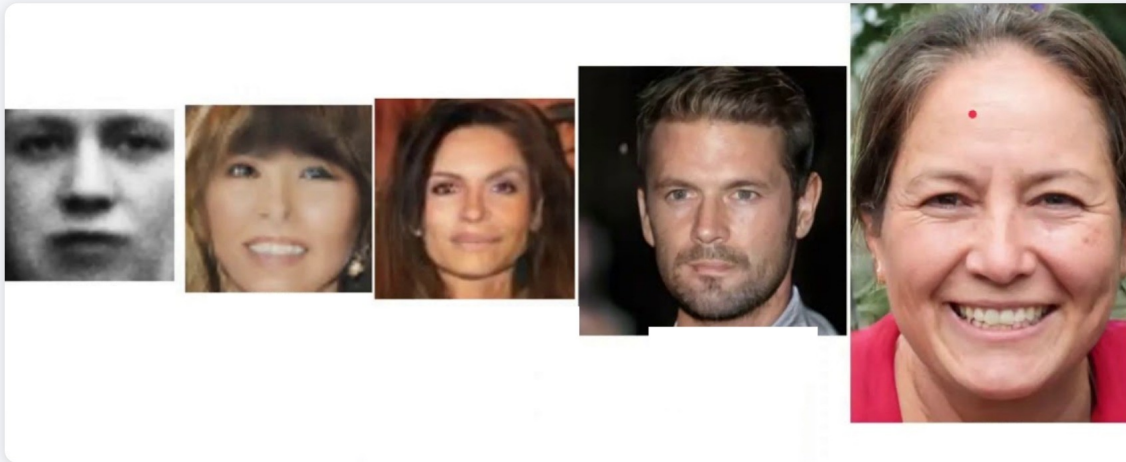
each condition lands on its own spiral

HOW THE GENERATORS WORK · 2 OF 2

Conditional WGAN-GP: generator versus critic

A generator and a Wasserstein critic compete until fake profiles are indistinguishable from real.

FAMILIAR · GANS FOR FACES



OURS · PROFILE GENERATOR

adversarial training: G vs critic



G trains until the critic
can no longer tell them apart

MODEL DEVELOPMENT

Building the RNA-seq generator

We compared how to process the data, handle study differences, train the model, and tell it what to generate.

01 Data scope

Sources

OSDR API

ARCHS4

Studies

Single

Multiple

Tissues

Per tissue

All tissues

02 Processing

Expression

Raw

CPM

TPM

Scaling

None

Z-score

MaxAbs

Features

All genes

HVGs

Reactome

L1000 map

03 Study effects

Alternative methods

None

Within-study z-score

ComBat / MBatch

MOBER

04 Model training

Generator

WGAN-GP

DDIM

Training source

OSDR only

ARCHS4 only

ARCHS4 then OSDR

05 Conditioning

Model inputs

Tissue

FLT / GC

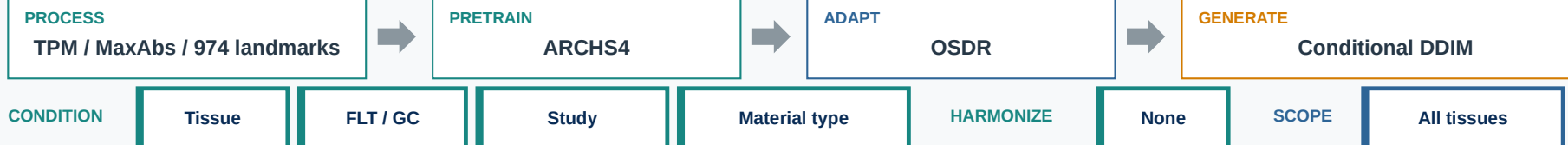
Study

Material type

Sex / age

Downstream

configuration

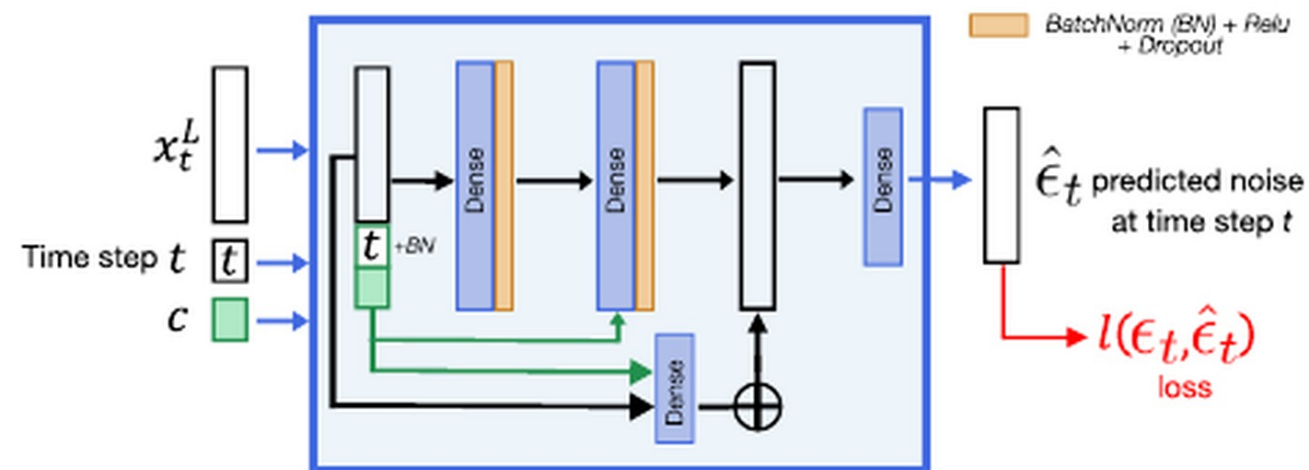


VALIDATION

Diffusion best reproduced the measured expression distribution

Both models reproduced gene expression. Diffusion samples were harder to distinguish from measured samples.

Architecture from Lacan et al.



The network repeatedly predicts and removes noise using tissue and time information.

For OSDR, the model also receives FLT/GC, study, and sample-material information.

Tissue identity retained

Real 0.781 | Synthetic 0.781

Balanced accuracy is preserved in generated tissue labels.

Metric	WGAN-GP	DDIM	Read
Expression corr.	0.976	0.974	higher
Coverage (F1)	0.985	0.997	higher
Real-vs-synth. acc.	0.636	0.475	0.5 is ideal
Distribution distance	0.144	0.074	lower

Use diffusion

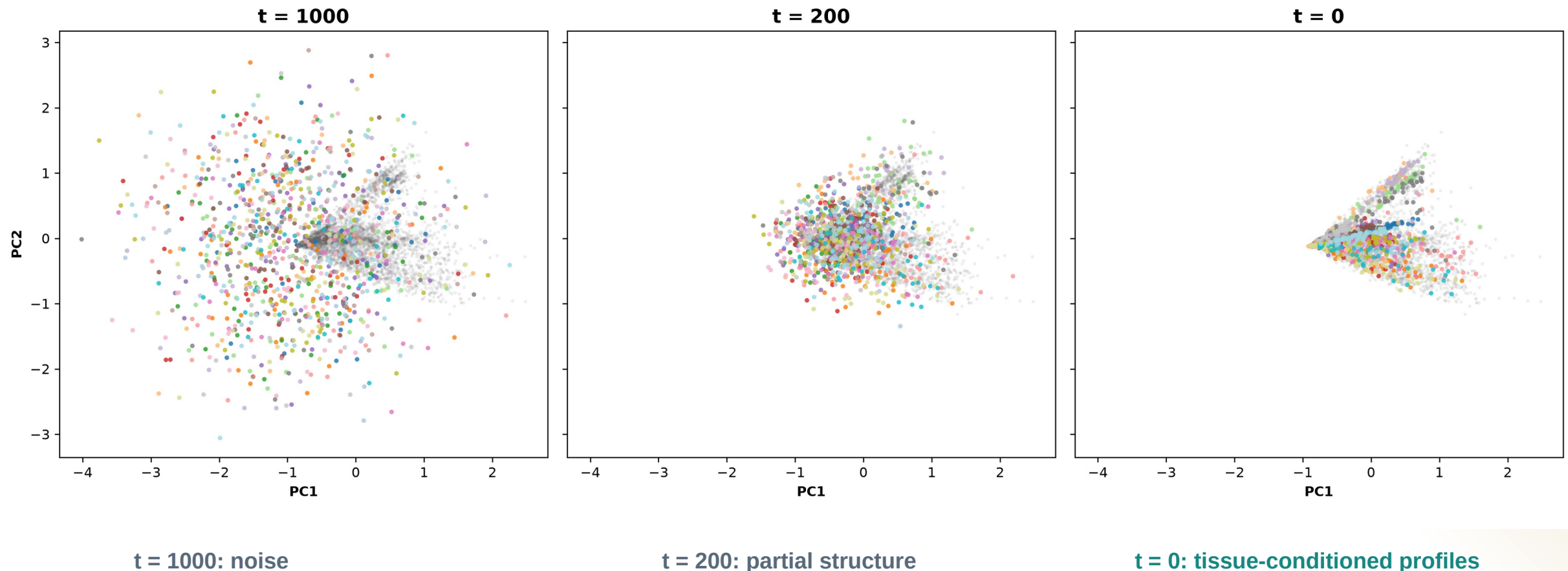
Better coverage and closer to the real distribution.

A score near 0.5 means a prediction model cannot reliably tell measured and generated samples apart.

DIFFUSION

Diffusion learns tissue structure from noise

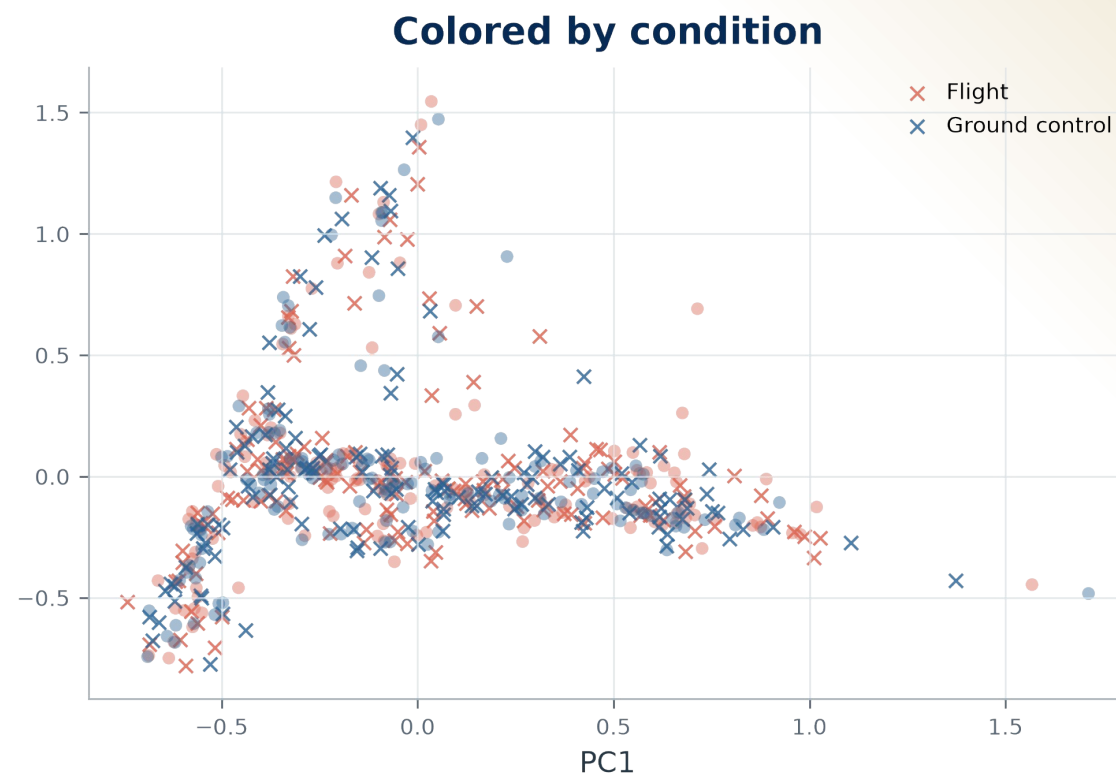
The same PCA axes follow 1,024 generated profiles through reverse diffusion.



DIFFUSION OUTPUT

FLT and GC overlap in the global PCA view

The condition signal is not clear when all tissues and studies are combined.

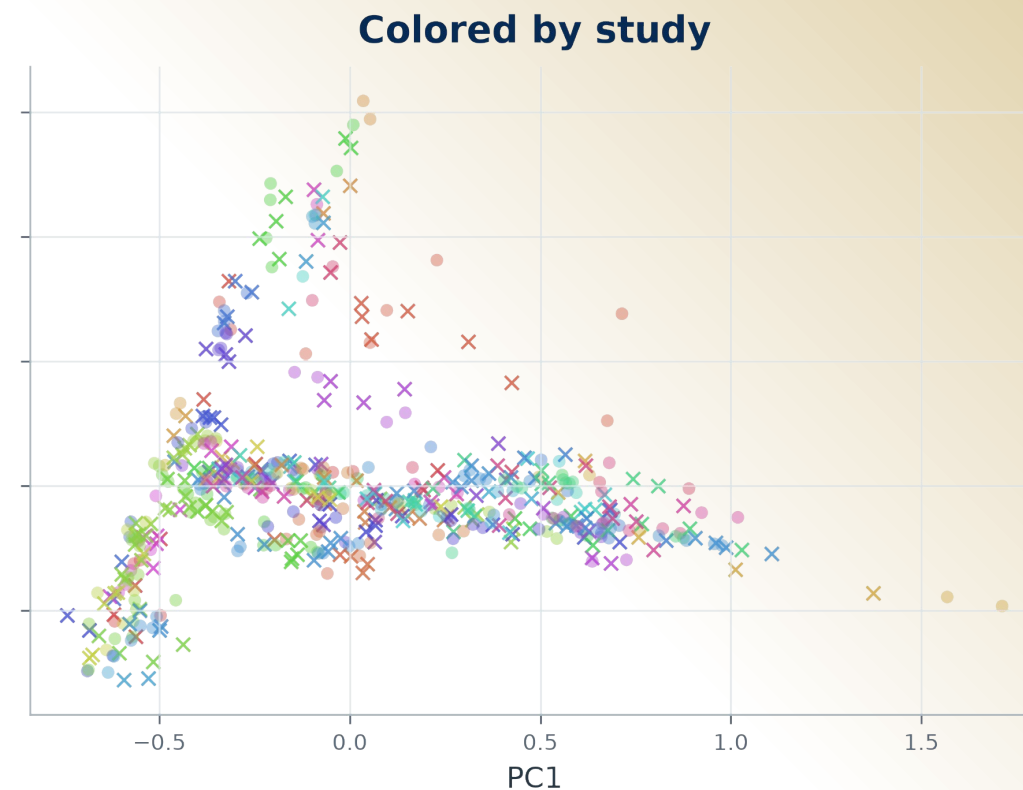
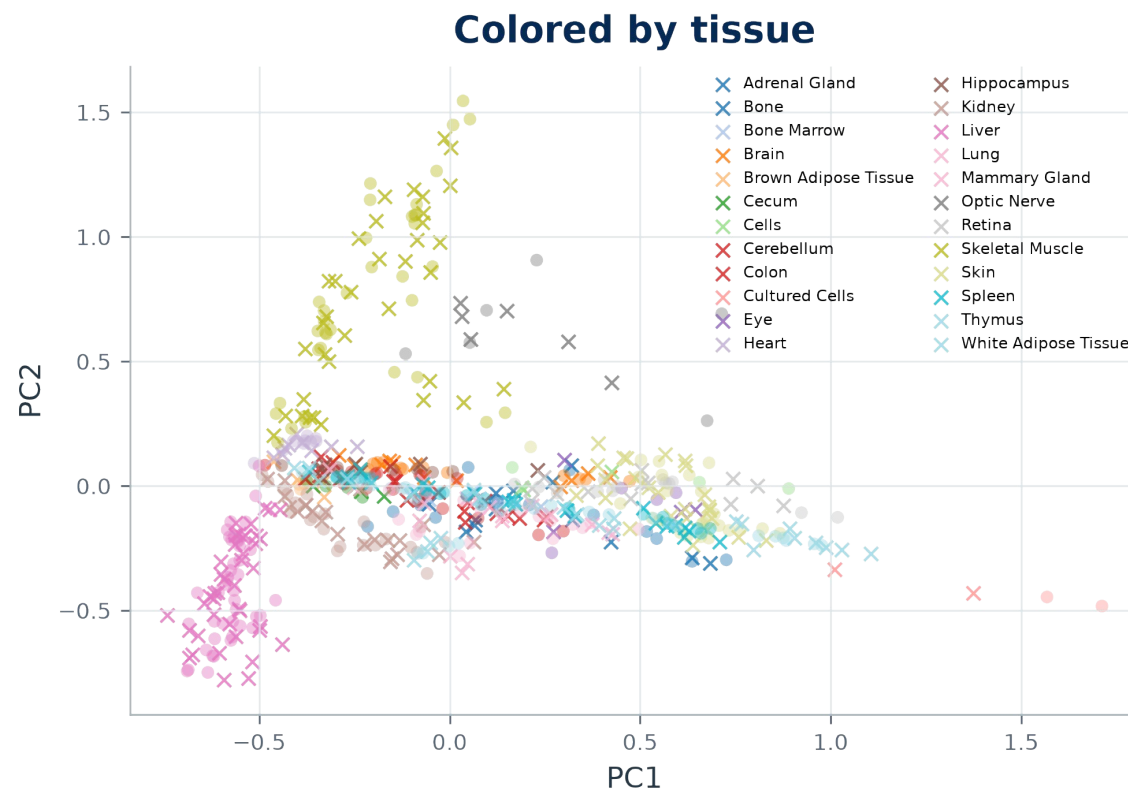


Color: FLT or GC. Circles are real samples; crosses are generated samples.

DIFFUSION OUTPUT

Tissue and study structure dominate the PCA space

We therefore compare FLT and GC within tissues and account for study.

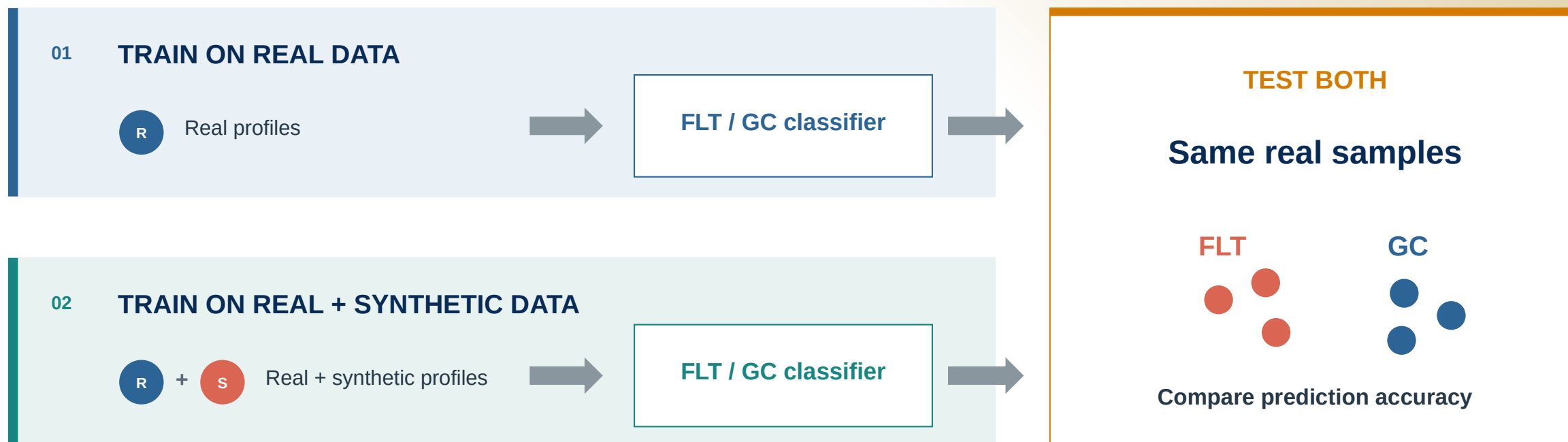


Left: tissue. Right: study. Circles are real samples; crosses are generated samples.

PRIMARY ANALYSIS

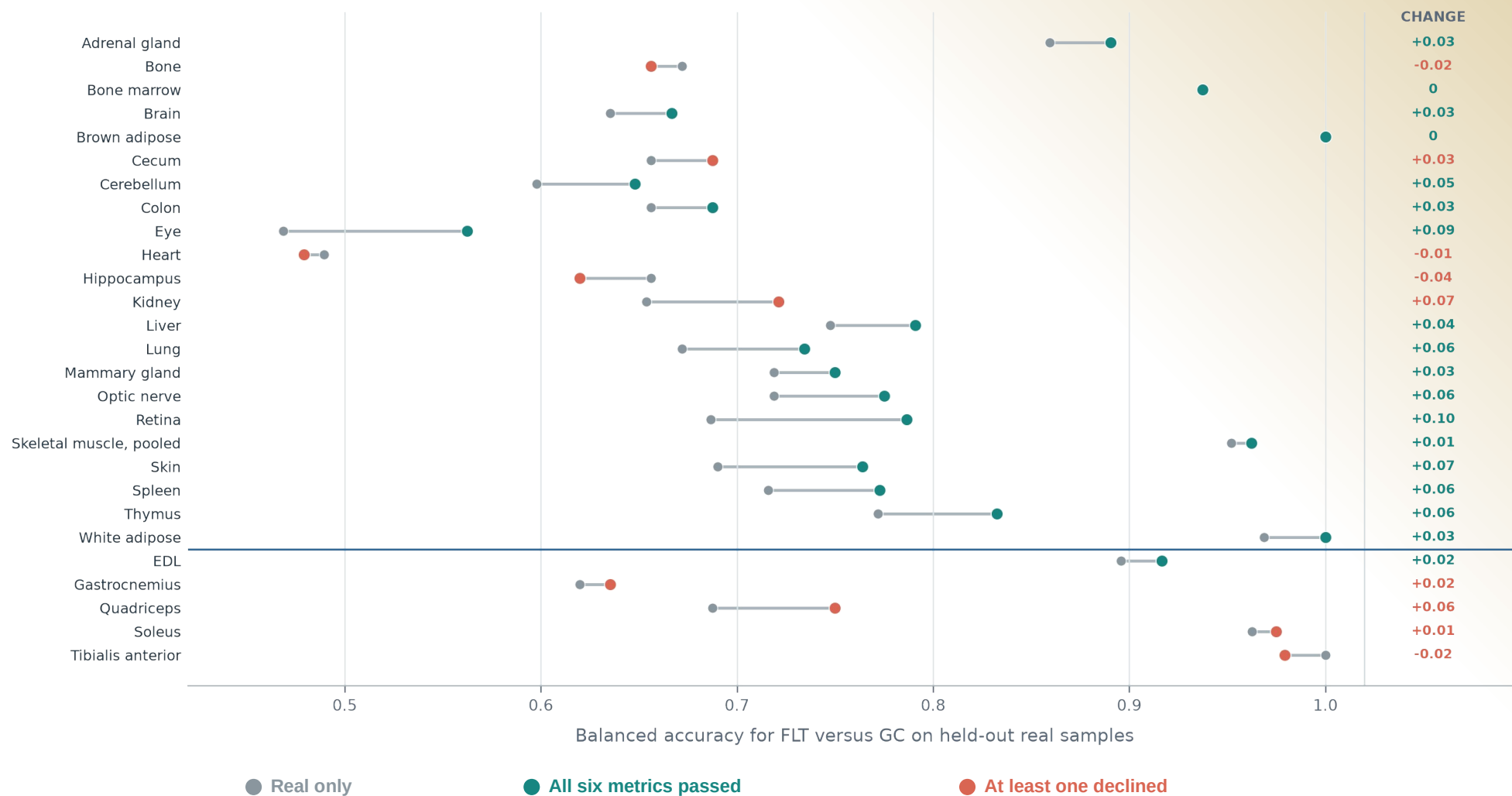
Does adding synthetic data improve FLT vs GC prediction?

Train the same tissue-specific classifier two ways, then compare both versions on real samples.

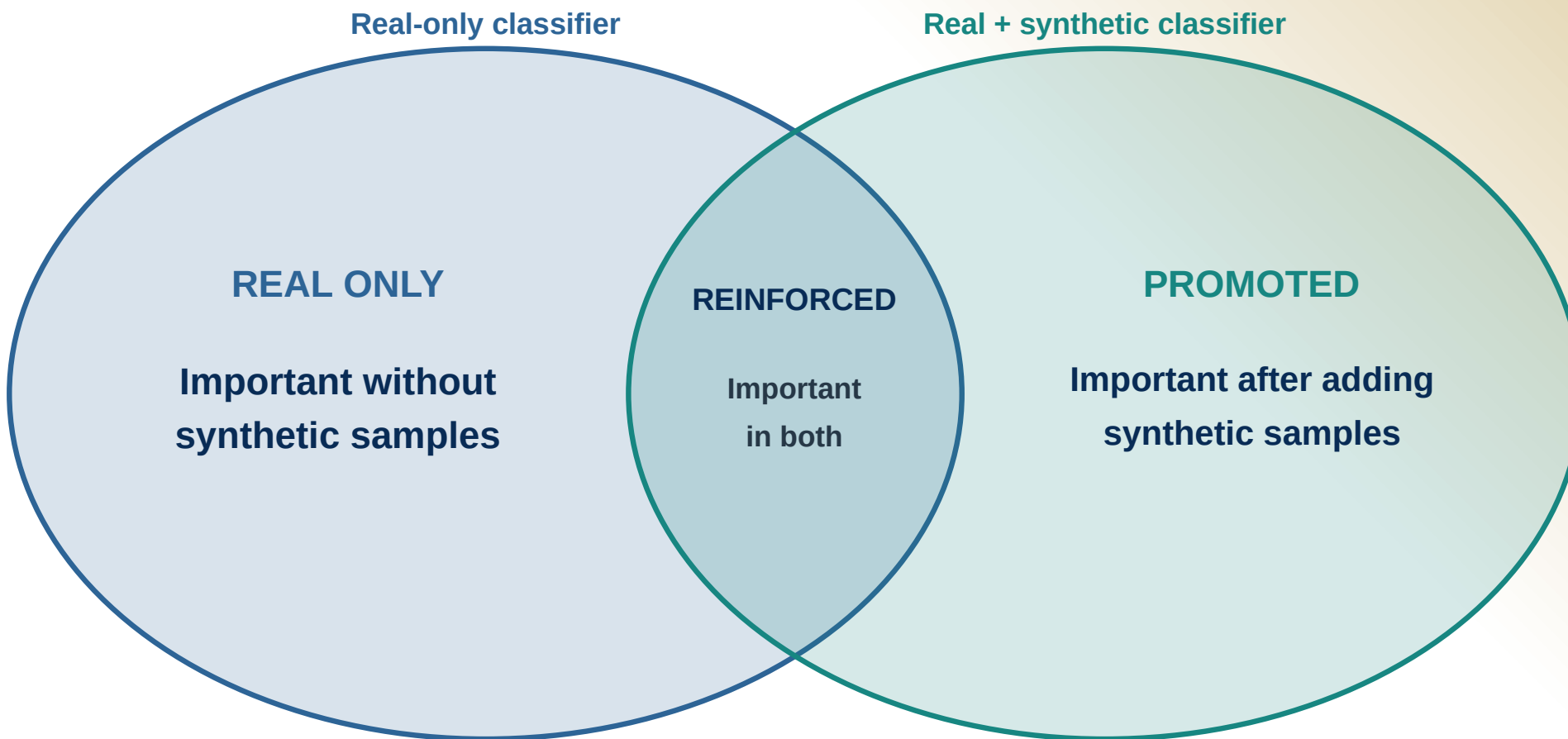


PREDICTION

Real + synthetic vs real-only balanced accuracy across 27 tissues



Compare what each classifier finds important



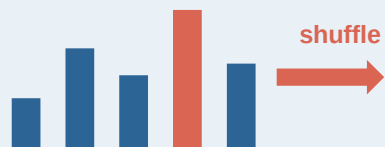
SYNTHETIC ANALYSIS

We compared feature importance at three levels

Each analysis compares the real-only and real + synthetic classifiers.

1 INDIVIDUAL GENES

Permutation + SHAP



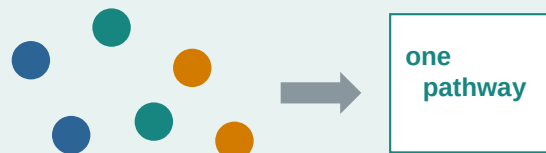
Permutation: does prediction weaken?

SHAP: does the gene push toward FLT or GC?

OUTPUT · RANKED GENES

2 REACTOME GROUPS

Grouped permutation + grouped SHAP



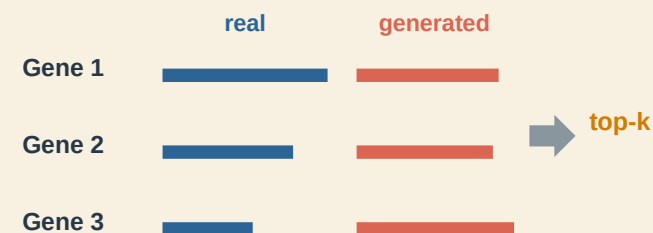
Move related genes together

Test correlated genes as one group

OUTPUT · RANKED PATHWAYS

3 CONSENSUS RANKING

Compact gene panels



Combine measured and generated ranks

Test small panels of the highest-ranked genes

OUTPUT · COMPACT GENE PANELS

RESULTS

The three analyses retained different types of results

Every retained candidate was also supported by the real OSDR data.

Individual genes

Permutation + SHAP

21 associations in 4 tissues

■ 7 promoted ■ 14 reinforced

RETAINED BY TISSUE

Thymus	15
Liver	4
Skin	1
Spleen	1

Reactome groups

Grouped permutation + SHAP

10 pathways in 3 tissues

■ 4 promoted ■ 6 reinforced

RETAINED BY TISSUE

Thymus	7
Skin	2
Spleen	1

Consensus ranking

Compact gene panels

49 associations in 10 analyses

■ 26 promoted ■ 23 reinforced

RETAINED BY TISSUE

Thymus	16
Pooled muscle	12
Soleus	5
Seven others	16

RESULTS

All three analyses found candidates in thymus, skin and spleen

TISSUE	Individual genes	Reactome groups	Consensus ranking
	Permutation + SHAP	Grouped permutation + SHAP	Compact gene panels
Thymus	15 7 promoted, 8 reinforced	7 2 promoted, 5 reinforced	16 13 promoted, 3 reinforced
Skin	1 0 promoted, 1 reinforced	2 2 promoted, 0 reinforced	1 1 promoted, 0 reinforced
Spleen	1 0 promoted, 1 reinforced	1 0 promoted, 1 reinforced	4 3 promoted, 1 reinforced
Liver	4 0 promoted, 4 reinforced	-	-
Pooled muscle	-	-	12 4 promoted, 8 reinforced
Soleus	-	-	5 0 promoted, 5 reinforced
Kidney	-	-	2 1 promoted, 1 reinforced
Tibialis anterior	-	-	4 1 promoted, 3 reinforced
Gastrocnemius	-	-	2 2 promoted, 0 reinforced
Adrenal gland	-	-	2 1 promoted, 1 reinforced
Eye	-	-	1 0 promoted, 1 reinforced

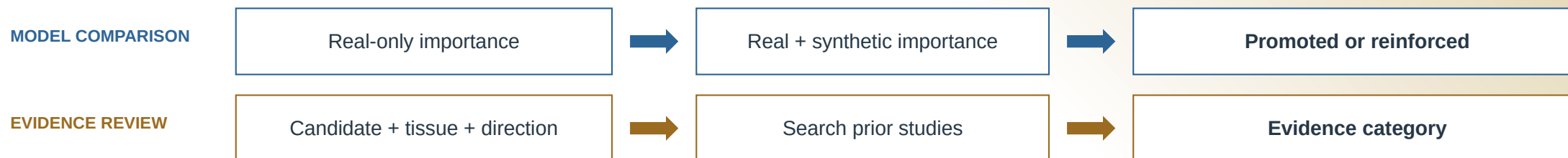
FOUND BY ALL THREE

Results below the divider came from one analysis.

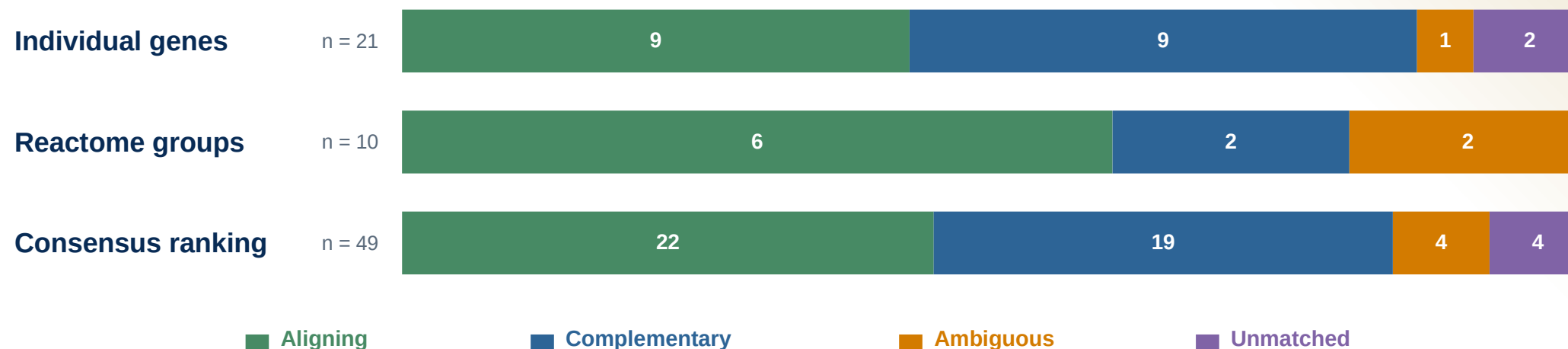
LITERATURE REVIEW

Feature selection and literature review answer separate questions

Selection compares classifiers. Literature review compares each candidate with prior evidence.



Literature classifications among retained candidates

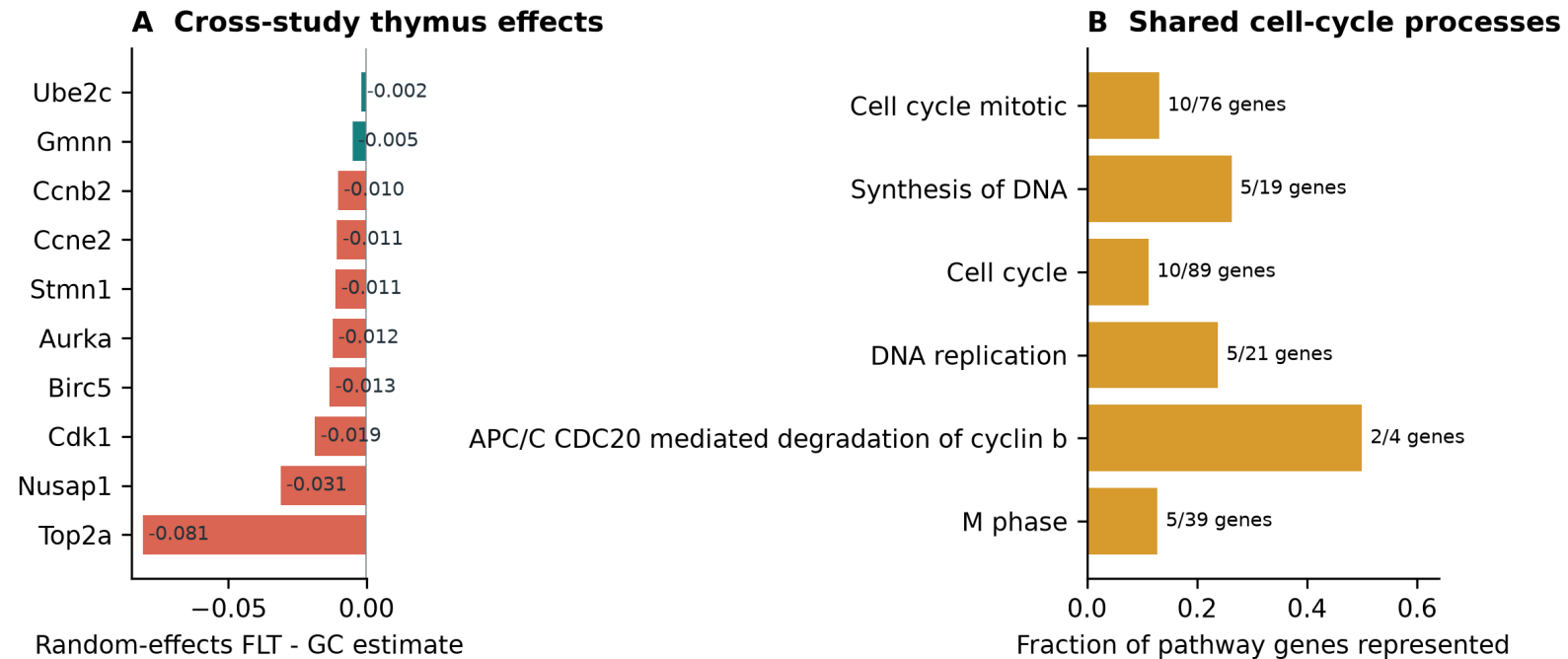


THYMUS

Thymus: all three analyses point to lower cell-cycle activity in flight

Genes, Reactome pathways and the compact panel all show the same direction.

Real OSDR effects and shared Reactome processes



INDIVIDUAL GENES

15**7 promoted | 8 reinforced**

Most retained genes are FLT lower.

REACTOME GROUPS

7**2 promoted | 5 reinforced**

Six pathways describe mitosis, DNA replication or APC/C control.

CONSENSUS RANKING

16**13 promoted | 3 reinforced**

Thirteen genes are FLT lower, including CDK1, TOP2A and AURKA.

BIOLOGICAL READING

Lower thymic proliferative activity

BIOLOGICAL INTERPRETATION

Skin and spleen show different flight-associated patterns

Skin: regulated cell death | Spleen: structural remodeling

SKIN

Interferon-linked regulated cell death

GENE EVIDENCE PLSCR1 was higher in flight and retained by two analyses

PATHWAY EVIDENCE Two necroptosis pathways were higher in flight

CONTEXT PLSCR1 is interferon inducible

COMPLEMENTARY HYPOTHESIS

Direct spaceflight evidence for skin necroptosis is limited

SPLEEN

Matrix, adhesion and cytoskeletal remodeling

GENE EVIDENCE LOXL1 was retained by two analyses; RAI14, PTPRK and MYL9 were also flight higher

PATHWAY EVIDENCE No shared structural Reactome pathway

CONTEXT The genes span matrix, adhesion and cytoskeletal functions

COMPLEMENTARY HYPOTHESIS

Matrix, adhesion and cytoskeletal remodeling

TAKEAWAYS

What the models can and cannot tell us

01 Computational evidence

expiMap and the generative models summarize patterns in the measured RNA-seq data. They do not reveal mechanism directly.

02 Candidates to follow up

Agreement among genes, pathways, models, and prior studies narrows the list of tissues and targets.

03 Experimental testing

Wet-lab experiments must confirm the biological changes and distinguish regulation from shifts in cell composition.

Thank you

Questions?

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Mentor

James Casaletto

Program

NASA Space Life Sciences Training Program

Data

NASA OSDR | ARCHS4 | Reactome

AI tools

ChatGPT | Claude

Code and
manuscript

github.com/jasont314/nasa-mouse

All biological associations were tested in observed OSDR profiles.